



Figure 5: Selenium WebLOAD configuration

feature, as well as connection throttling.

- Grinder has a sophisticated feature to record and replay the interaction between a browser and a website.

Selenium

Selenium (<https://www.seleniumhq.org/>) is a suite of tools to automate Web browsers across many platforms. It runs in many browsers and operating systems, and can be controlled by many programming languages and testing frameworks. Figure 6 shows the configuration for carrying out Selenium based load testing.

There are many other tools, as listed below, which also can be considered as alternatives for performance and load testing, based on an organisation's needs.

Locust.io: This is a great tool that is mainly used for understanding the performance on the server side.

Bees with Machine Guns: This is a utility based tool for creating micro EC2 instances to attack Web applications as targets.

Multi-Mechanize: This is an open source framework for performance and load testing that runs concurrent Python scripts to generate load (synthetic transactions).

Siege: This HTTP load-testing and benchmarking utility was designed to let Web developers measure code under duress, to see how it stands up to load on the Internet. Siege supports basic authentication, cookies, HTTP and HTTPS protocols, and lets the user hit a Web server with a configurable number of simulated Web browsers.

Apache Bench: This tool is used for benchmarking an Apache HTTP server,

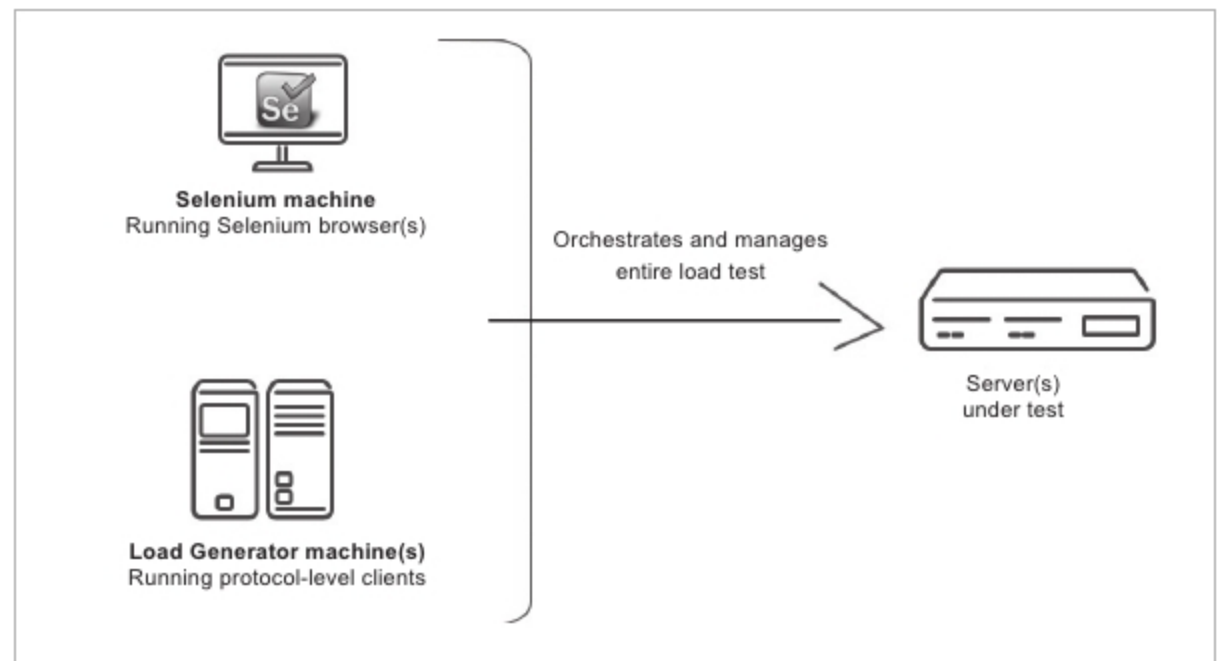


Figure 6: Load testing using Selenium (Source: <https://speedcurve.com/blog/web-performance-page-bloat/>)

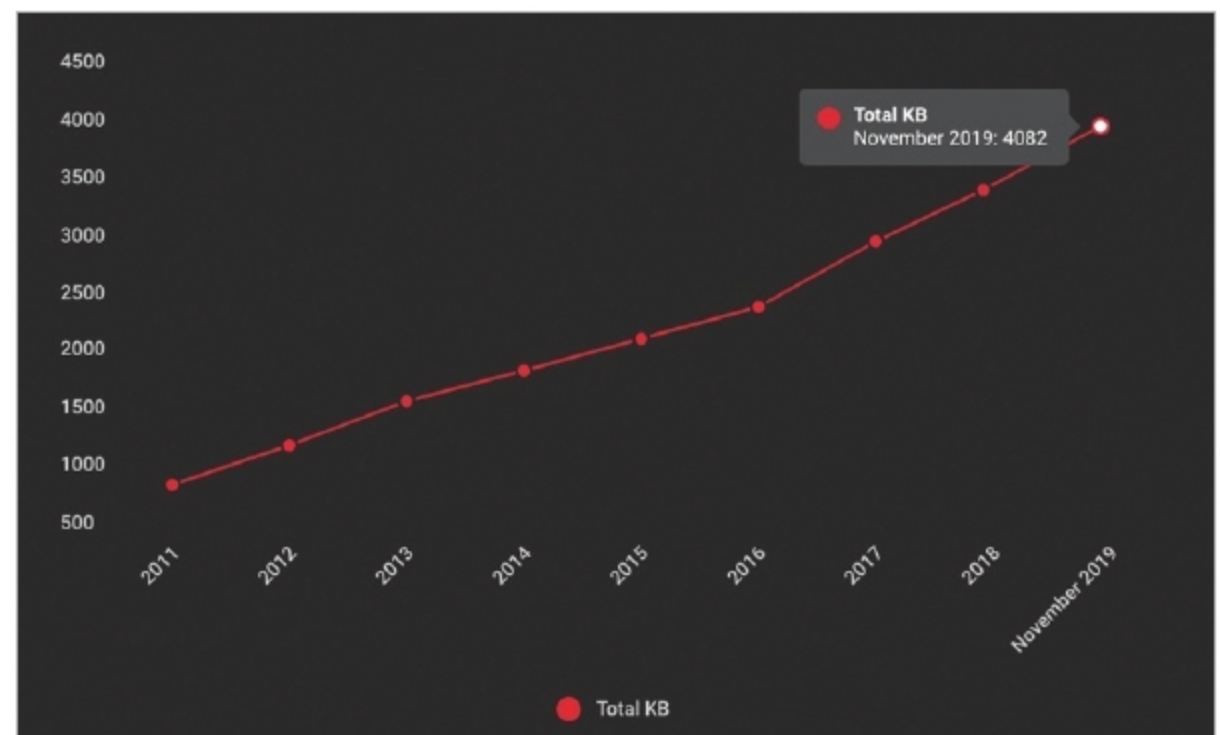


Figure 7: Measuring the growth rate of a 4MB Web page (Source: <https://speedcurve.com/blog/web-performance-page-bloat/>)

to get an evaluation on how it performs.

Httpperf: This tool measures a Web server's performance and provides a flexible facility for generating varied HTTP workloads and measuring server performance. The focus is not on implementing a particular benchmark but on providing a robust, high-performance tool that facilitates the creation of both micro- and macro-level benchmarks.

Web page performance testing for scalability, load and stress needs to be carried out on a constant basis to make sure that all needs of the future are met. Figure 7 shows the rate of growth of a 4MB Web page. Once the test results are out, some of the above mentioned optimisation methods and measures can be taken for future performance enhancement of Web pages and applications. **END** 🐧

By: Shashidhar Soppin

The author is a senior architect with 18+ years of experience in the IT industry. He specialises in virtualisation, Docker, the cloud, AI-ML, deep learning and OpenStack. He has patents to his credit and has written many papers. You can contact him at shashi.soppin@gmail.com.